

Team Details

a. Team name: Hey Merlion!
b. Team leader name: Changhyeon Nam (team: Changhyeon Nam, Hyunho Yang)
c. Problem Statement: BFSI: Intelligent Risk, Fraud & Financial Experiences

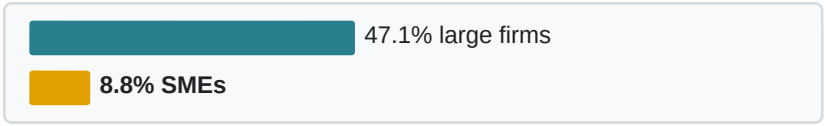
Why this matters

SOUTHEAST ASIA'S SMALL BUSINESSES RUN THE ECONOMY BUT CANNOT BORROW

8.8%

of Vietnam's 914,000 SMEs have a bank loan.

Large enterprises: 47.1%. The same small firms employ most of the region.



75.5% of Vietnamese firms cannot borrow without collateral. What they have is PDFs, screenshots and paper receipts.

70% of emerging-market MSMEs lack adequate financing. Reading their records by hand costs more than a small loan earns.

The opening VPBank, Vietcombank and Agribank are moving from collateral to cash-flow lending. Someone still has to read the cash flow. LedgerLens makes that reading fast, sourced and auditable.

Sources: FiinGroup via Viet Nam News (Aug 2026) · VCCI Private Economic Report 2025 · IFC MSME Finance · VTV (May 2026) · ADB Asia SME Monitor 2025

Brief about the idea

LedgerLens turns a shoebox of receipts, screenshots and bank PDFs into a credit memo a bank can trust.

1 Read with evidence

Gemini reads every document. Every value carries document id, page and bounding box. Python computes the cash flow. A memo with unsourced numbers is rejected and regenerated.

2 Challenge

A second agent that never sees the first one's prompt runs 12 underwriting checks against the memo and raises severity-ranked objections with evidence.

3 Human signs

The underwriter accepts or dismisses each objection and signs. Every decision is logged and re-weights the next Challenger run. The applicant gets the reasons in their own language.

Not a score. A memo. AI does the reading and the arguing; the underwriter keeps the decision.

Opportunities

Same documents. About 2 minutes per file. Every number defensible.

~2 min

per file, end to end (measured, live Gemini)

~0.28

USD per case on a cold run; USD 0 on a cached re-run

0 vs 62

unsourced numbers in the memo: LedgerLens vs a single prompt

How it is different

- Writer and verifier are separate agents, so the memo is never graded by its author
- Figures are computed by code and linked to their source line, not generated
- Underwriter decisions re-rank the next Challenger run

Who benefits

- Small businesses with records but no credit file get assessed on cash flow, not collateral
- A declined applicant receives the reasons, in their own language
- Lenders can offer small-ticket loans that were uneconomic to underwrite

List of features offered by the solution

Every feature hangs on one pipeline, and the pipeline ends with a person.

Intake → Extract → Cash flow → Analyze → **Challenge** → Decide → Explain

Multimodal intake

PDF, screenshots, photos. Classified and quality-flagged.

Evidence-linked extraction

Doc id, page, bbox on every value. PDF text layer verifies it.

Deterministic model

Inflow, outflow, trend, volatility, debt service. Python, not the model.

Clickable memo

Cites catalog figures only. Click one, the source line opens.

Battery + 12 checks

Image hash, balance chain, dates. Ranked objections with evidence.

Underwriter workbench

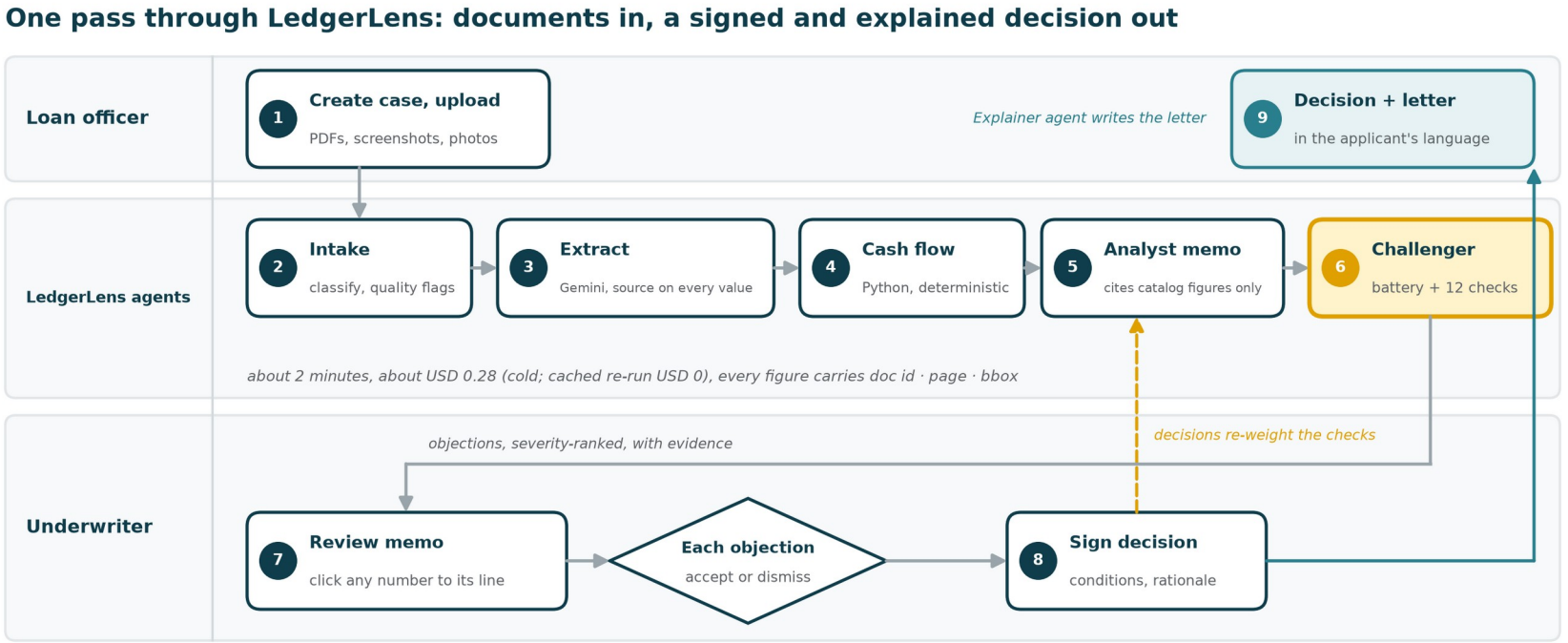
Accept or dismiss, sign. Audit log. Decisions re-weight the checks.

Letter and paper memo

Customer letter in the local language. Memo prints with footnoted sources.

Out of scope by design: credit scoring, loan pricing, core-banking integration. LedgerLens informs the decision; it does not make it.

Process flow diagram

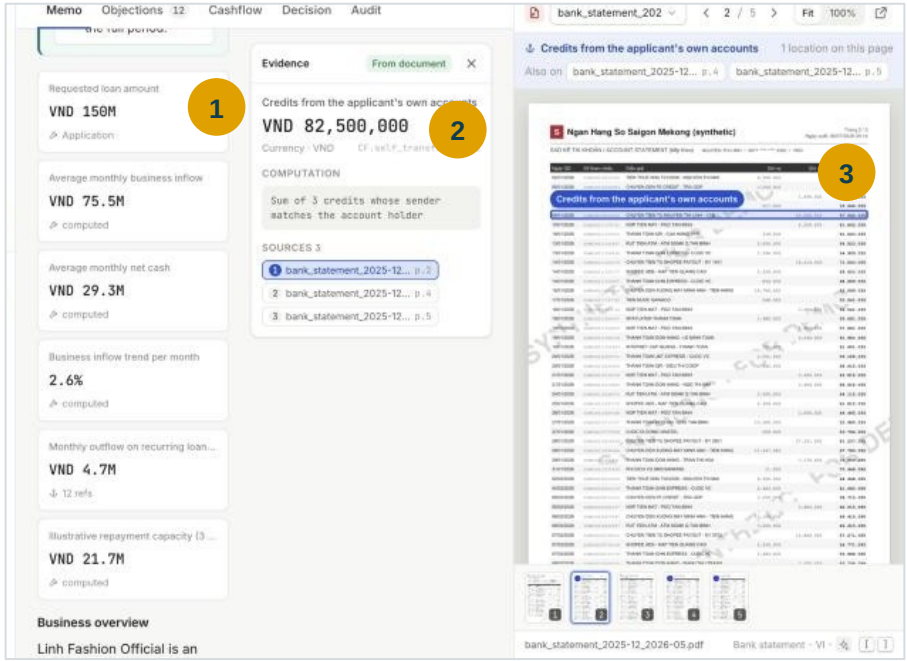


Three lanes: the loan officer uploads, the agents read and challenge, the underwriter decides. Decisions flow back into the Challenger's priorities.

Wireframes / Mock diagrams of the proposed solution

THE PRODUCT IN ONE INTERACTION

Click the number. See the line.



1 The memo cites a figure

Every underlined value is a catalog figure with a source. Here: VND 82,500,000, credits from the applicant's own accounts.

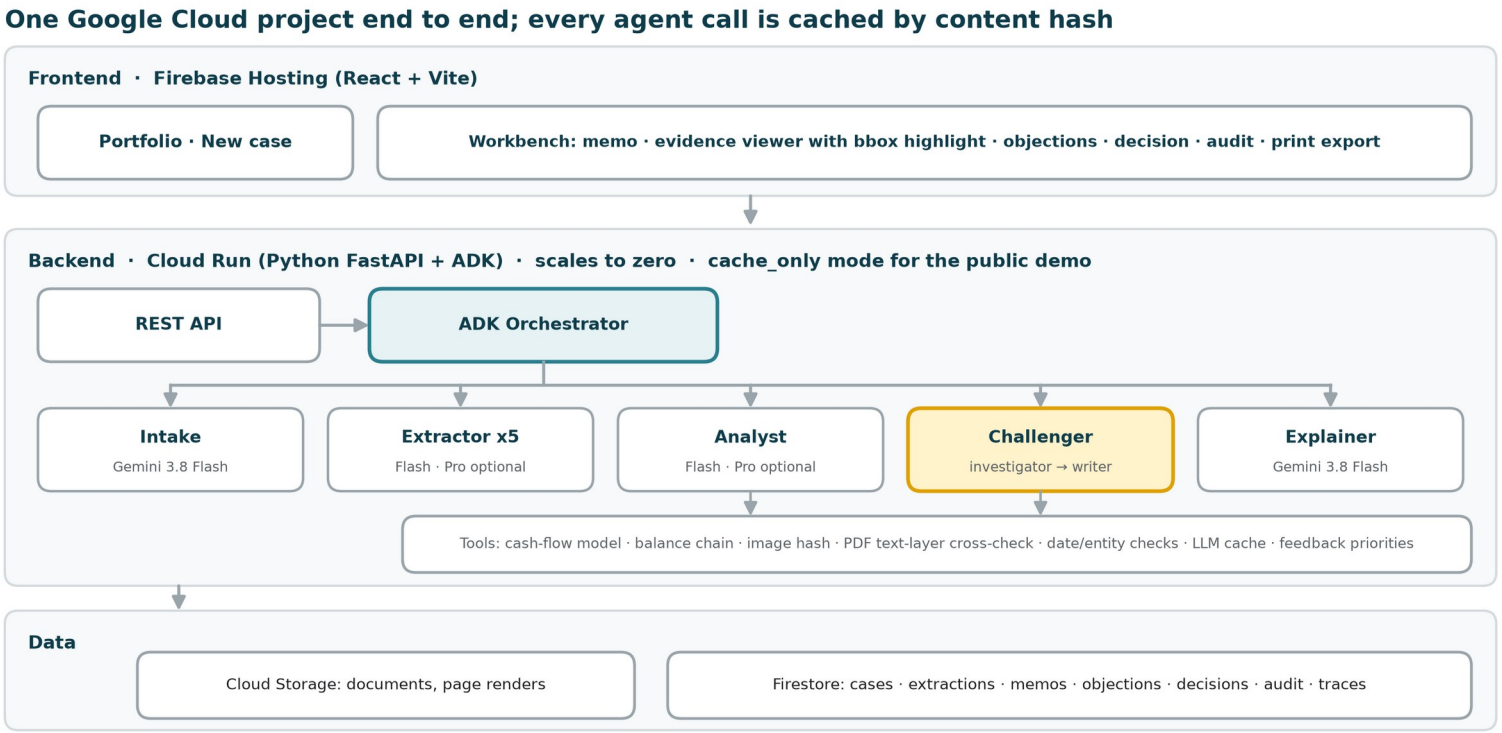
2 One click opens the evidence

The computation (sum of 3 credits whose sender matches the account holder) and all 3 source references: pages 2, 4 and 5.

3 The source line lights up

The statement page opens with the extractor's bounding box drawn on the exact row. A memo with an unsourced number is rejected and regenerated.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Gemini reads, ADK orchestrates, Python computes, Cloud Run serves.

GOOGLE CLOUD

Gemini 3.8 Flash

Default for every agent. Measured USD 0.26-0.31 per cold case; USD 0 cached.

GOOGLE CLOUD

Gemini 3.1 Pro

One env var moves Extractor, Analyst and Challenger to Pro.

GOOGLE CLOUD

Agent Development Kit

Intake, 5 Extractors, Analyst, Challenger (investigator and writer), Explainer.

GOOGLE CLOUD

Cloud Run + Firebase

FastAPI backend scales to zero; React workbench on Firebase Hosting.

GOOGLE CLOUD

Firestore + Storage

Cases, extractions, memos, objections, decisions, audit, traces; documents.

PYTHON

Deterministic tools

Cash-flow model, balance chain, image hash, PDF text-layer cross-check, date and entity checks.

PYTHON

LLM cache, budget cap

Content-hash cache per model and prompt version; hard USD cap per process.

INTEGRATION

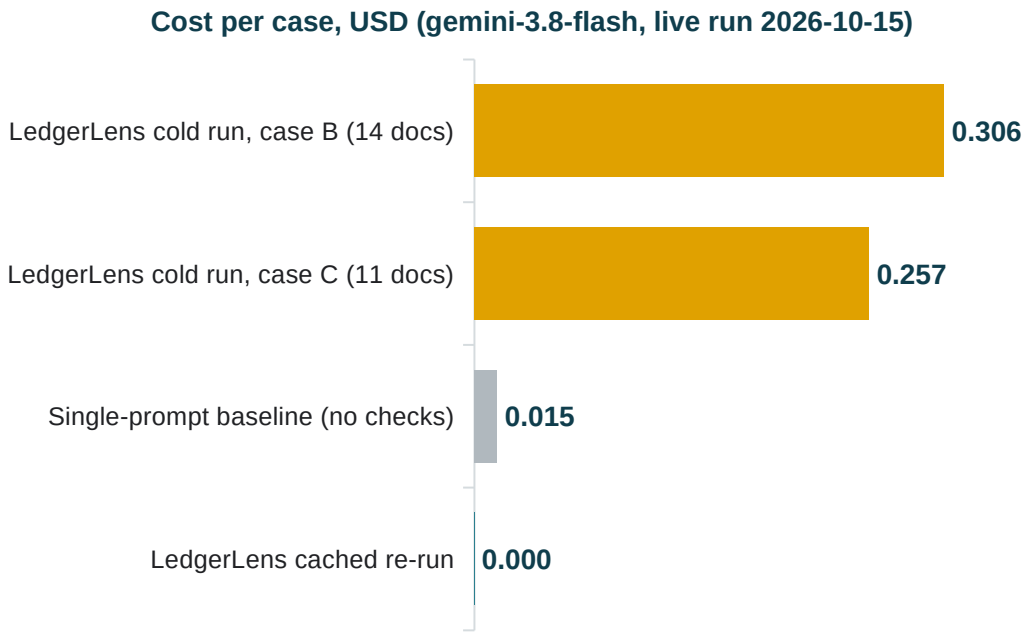
REST API

Any loan origination system can create cases, run stages and read memos.

Output schema requires source {doc_id, page, bbox} on every value. No separate OCR engine: pages are rendered to images and read by Gemini.

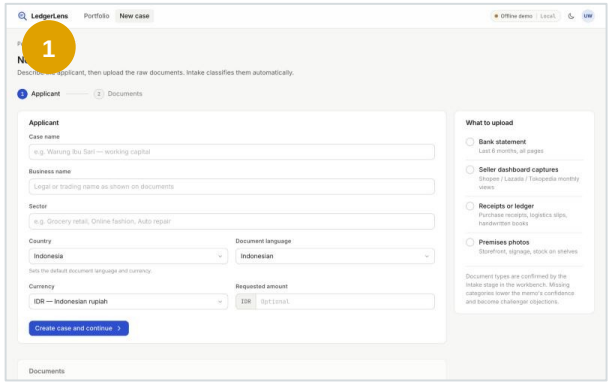
Estimated implementation cost

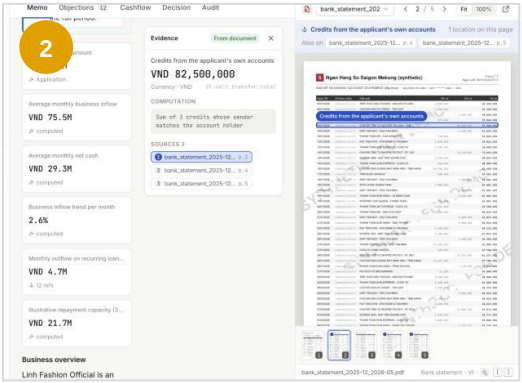
Measured: USD 0.26-0.31 per case on a cold Flash run, USD 0 on a cached re-run

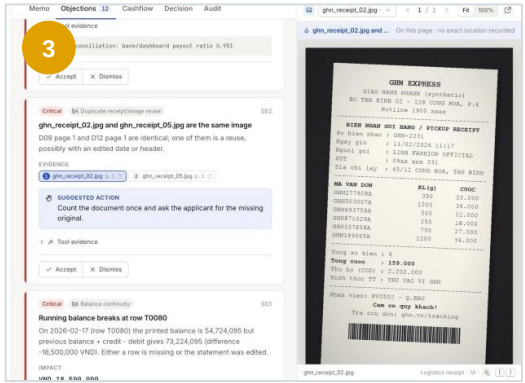


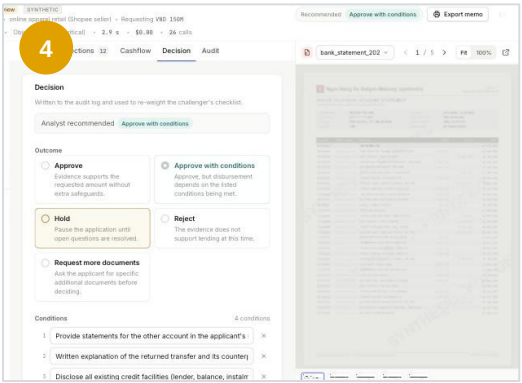
Assumption	Value
Gemini 3.8 Flash list price	USD 0.75 / 3.75 per 1M tokens
Measured cold run	~2 min and USD 0.26-0.31 per case
Cached re-run	USD 0 (content-hash LLM cache)
Public demo	Replays the recorded live Gemini run, USD 0; only the customer letter calls Gemini (~USD 0.0015)
Hosting	Cloud Run scales to zero; Firestore and Storage in free tier
Guardrail	Budget-cap env var stops Gemini calls once spent
Build	2 engineers, 4 days, free tier and trial credits

Snapshots of the prototype









1 New case

Applicant details, then the raw documents. Intake classifies and quality-flags them.

2 Click a number, see the line

Evidence panel with computation and all sources; the statement opens with the bounding box highlighted.

3 Challenger objections

Critical: two receipt photos are the same image with an edited date. Evidence chips open the photo.

4 The underwriter signs

Outcome, conditions and rationale, written to the audit log; accept/dismiss rates re-weight the checks.

Prototype performance report / Benchmarking

18 of 20 planted traps caught on live Gemini output; a single prompt on the same documents caught 3.

LEDGERLENS

18

of 20 traps caught

95% Wilson CI 0.70-0.97

SINGLE PROMPT

3

of 20, same documents

and same model

0

unsourced numbers in the memo (single prompt: 62)

~2 min

per case cold, USD 0.26-0.31; cached re-run USD 0

Metric	Target	Result
Trap recall (strict)	≥ 16 / 20	18 / 20
Unsourced numbers	0	0 (single prompt: 62)
Objections on clean case A	≤ 20%	0
Extraction row accuracy*	≥ 97%	100% (481 / 481)
Evidence-link accuracy*	≥ 95%	100%
Time per case (cold)	≤ 180 s	~2 min
Cost per case (cold)	≤ USD 0.50	USD 0.26-0.31

How measured. Live gemini-3.8-flash runs, 2026-10-15/16: 3 synthetic businesses, 43 documents, 20 planted traps. Strict scoring: an objection counts only if it carries the trap's identifying amount or document, one objection per trap. **Not a blind evaluation:** these three cases informed the fixes (demo watermark handling, ledger/deposit alignment, weekly-lender detection, recommendation rule); held-out cases are future work. n = 3. Missed: dashboard gross vs payout, COD vs cash deposits. Recommendation matched on C; A gave approve_with_conditions (expected approve), B request_more_documents (expected approve_with_conditions). *Clean born-digital synthetic PDFs; real scans will be lower.

Additional details / Future development

Where this goes next, and what we do not claim yet

Pilot Q1 2027
One SEA bank or MFI, real consented SME files; underwriter decisions replace synthetic traps

→

Integrate Q2–Q3 2027
Open-banking feeds (Brankas, Finverse) beside PDFs; hooks into loan origination systems

→

Scale 2028
Same pattern for trade finance and micro-insurance; shared Challenger check library

What we do not claim yet, and which phase answers it

Synthetic benchmark
3 synthetic businesses (n = 3) that also informed fixes; held-out and real-data cases come with the pilot.

Forged documents
Hash and balance-chain checks catch crude edits. Open-banking data closes the gap for the rest.

Platform sellers
Shopee and Grab already lend to their own sellers. We target banks and MFIs lending outside platforms.

positioning, not a feature

Humans decide, every figure and decision is traceable, the prototype uses watermarked synthetic data only, and production keeps documents and model endpoints in-region.

Provide links to your:

GitHub public repository

github.com/changhyeonnam/ghack

English README, architecture, run instructions, synthetic-data notice, presentation/

Demo video (up to 3 minutes)

TBD

Problem in 20 seconds, click-to-source and Challenger scenes, benchmark near the end

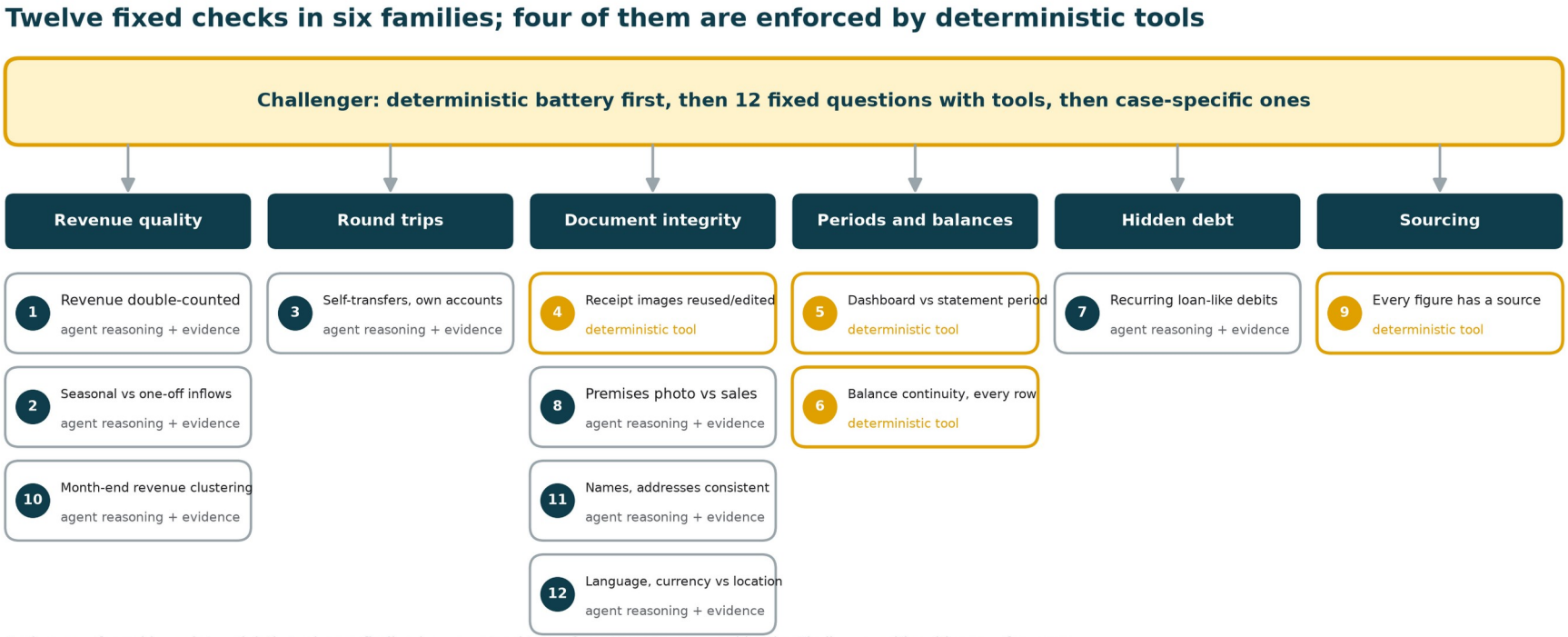
Final product

TBD

Cloud Run service; each browser gets its own copy of the three demo cases (Reset demo). Scales to zero; demo cases replay the recorded live Gemini run (only the customer letter calls Gemini live), so judging costs nothing

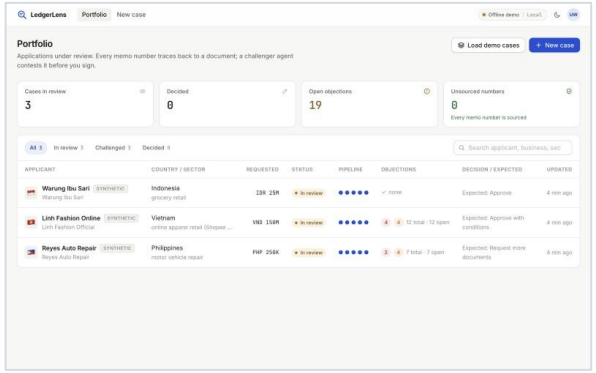
Links are inserted at submission on 18 October 2026.

Appendix: the Challenger's 12 fixed checks



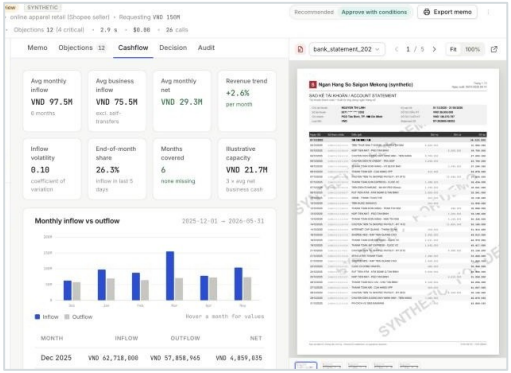
Amber = enforced by a deterministic tool, so a finding is guaranteed to surface. Grey = reasoned by the Challenger with evidence references.

Appendix: portfolio, cash flow and audit views



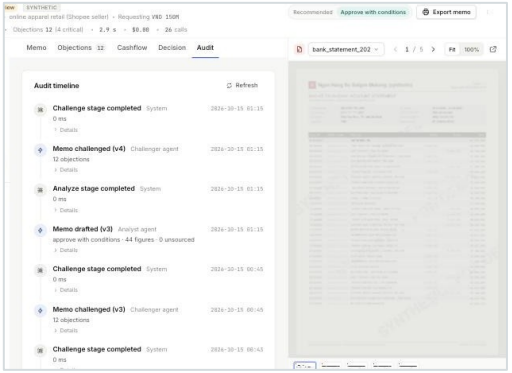
Portfolio

Cases in review, open objections and unsourced numbers across the book; filters by status.



Cash flow

Deterministic monthly model: inflow, net, trend, volatility, end-of-month share, repayment capacity.



Audit timeline

Every stage and agent call with version, model and prompt; the trail a regulator can read.

Also in the build: dark and light themes, PDF text-layer cross-check on every extracted amount, content-hash LLM cache, cache_only demo mode, budget cap, printable paper memo with footnoted sources.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

